



KJ'S EDUCATIONAL INSTITUTES
TRINITY COLLEGE OF ENGINEERING AND RESEARCH
(Accredited by NAAC with A+ Grade Approved by AICTE & Affiliated to SPPU, Pune)

Project Title:

***Continuous attention monitoring in vc
using media pipe and CNN***

Team Members:

Siddhant Babar (CO3124)

Tejas Mandhare (CO3146)

Nikhil Madle (CO3147)

Omkar Takawane(CO3109)

Project Guide:

Prof.Prasad Bhosale

Problem Statement

- In online meetings and classrooms, it is difficult to monitor participant attention.
- Manual observation is inefficient.
- Maintaining attention during video conferencing is challenging, as existing systems cannot detect whether users are attentive, distracted, or drowsy.

Objective

- Detect attention using AI.
- Identify distraction and inactivity.
- Send alerts to user and host.
- Improve engagement.

Technologies Used

- OpenCV (For video capture and image processing)
- Deep Learning (CNN)
- MediaPipe (For Facial landmark detection)
- LDTN
- Flask / Web App
- Database

Working Methodology

- Capture video frames
- Face detection
- Eye tracking & head movement
- Attention classification
- Alert system

Research Paper 1 Review

Year	Author	Title	Methodology	Advantages	Disadvantages
2024	Shreeya Patil Aayushi jani	Student's Attention and Alertness Monitoring System using AI	Face Detection: Our attention monitoring approach is made up of a series of linked steps intended to measure attention levels in real time. Using the cv2.CascadeClassifier class, a pre-trained Haar Cascade classifier is first loaded in order to identify faces. Then, each frame of the video is subjected to the detect Multiscale method, and the resulting coordinates (x, y, width, and height) of faces that are detected are obtained. Next, we use a shape predictor from the dlib library to predict facial landmarks	1) Real time Monitoring 2) Uses simple and efficient methods. 3) Gives useful feedback	1) Accuracy Depends on conditions. 2) Limited to one face

Research Paper 2 Review

Year	Author	Title	Methodology	Advantage	Disadvantage
2023	1)Perumal B 2)Nagaraj P 3)Thulasi Sai Narsimha Charan	Student Engagement Detection in Classroom using Deep CNN-based Learning Approach	The methodology of this paper involves collecting a dataset of around 600–1000 student images with different head poses and eye movements, which are labeled as <i>engaged</i> or <i>not engaged</i> . These images are then pre-processed through resizing and normalization to make them suitable for model training. The dataset is split into training and testing sets, and two deep learning models—CNN and MobileNet—are developed and trained on the data. The CNN model extracts features using convolution and pooling layers, while MobileNet provides a lightweight architecture for efficient processing. After training and validation, the system performs classification to predict whether a student is engaged or not based on input images.	1) Uses deep learning models (CNN & MobileNet) which provide high accuracy in image-based classification tasks.	1) Requires a large labeled dataset, which can be time-consuming to collect and annotate.

Features

- Real-time monitoring
- Attention scoring
- 3-time warning system
- Host notification
- Analytics dashboard

Expected Outcome

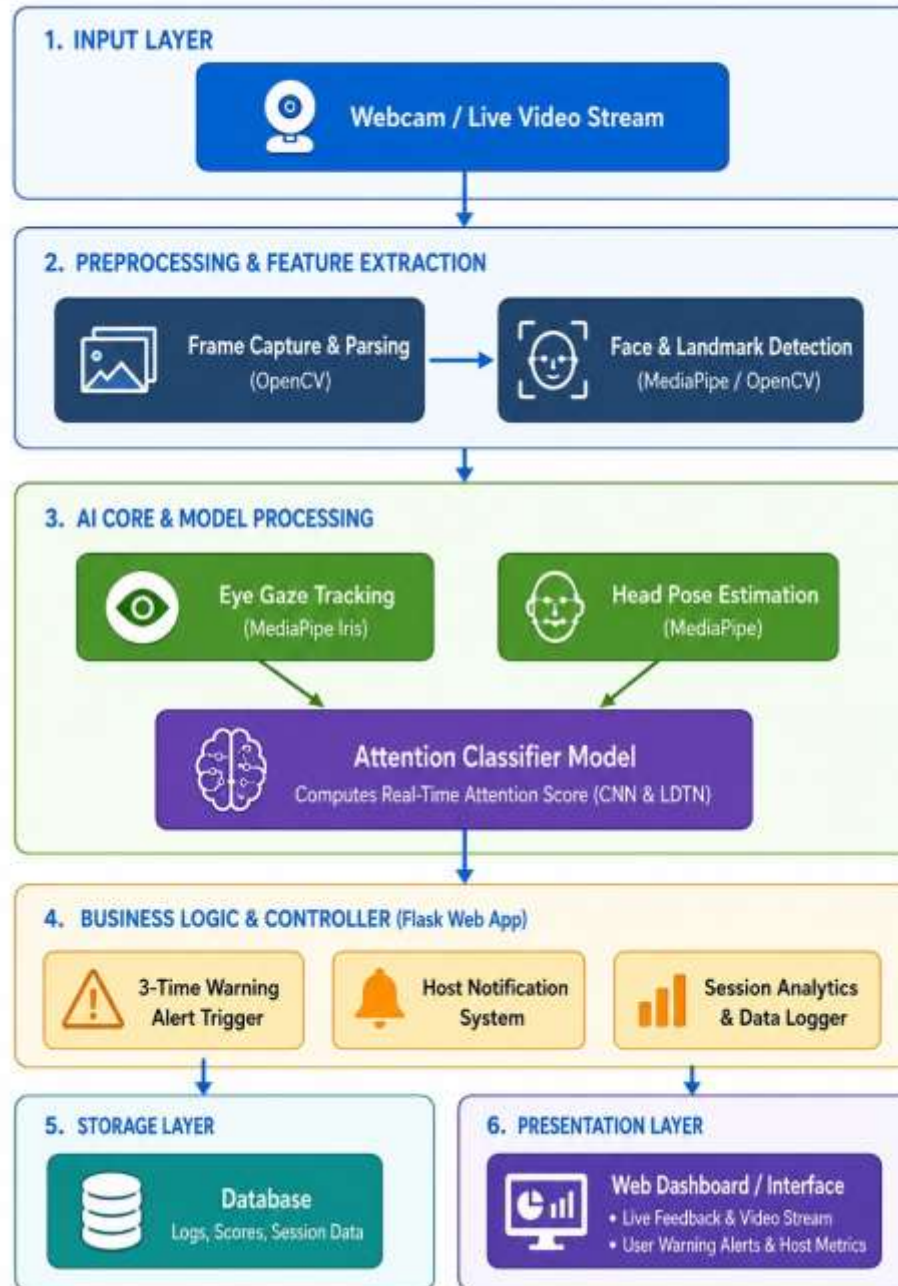
- Automated monitoring
- Attention Score Generation
- Better engagement
- Automated alert system

Applications

- Online classes
- Video conferencing
- Employee monitoring
- Smart classrooms

Person Attention Monitoring System

System Architecture Diagram



Advantages & Limitations

- **Advantages:**
 - Real-time Monitoring
 - Automation
 - High Accuracy (Mediapipe + CNN)
 - Detects Multiple Behaviors
 - Continuous Monitoring over time
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- **Limitations:**
 - Lighting dependency

Thank You